

Thermodynamic Properties of the Internal Combustion Engine

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1 Introduction

Our aim is to investigate the relative effectiveness of three different heat transfer mechanisms subject to different environmental conditions. In an internal combustion engine (ICE), fuel is ignited and combusted within the engine cylinder which acts as the method of heat input into the system. This creates a force against the piston which forms a cyclic system where the motion rotates a crankshaft (V.T.O. 2013). In our case, the crankshaft motion was used in an electric power generator to produce electricity. We used three ICEs in our experiment. The first two engines are petrol engines using spark ignition¹. The four-stroke ideal cycle for spark ignition is attached in the Appendix in Figure 5. We used a full throttle and half throttle engine². The third ICE we used was a diesel engine at full throttle, which uses compression ignition³. The four-stroke ideal cycle for compression ignition is attached in the Appendix in Figure 6. An image of our apparatus is also pictured in Figure 7.

For each of the three engines we made measurements at six different torque intervals of the engine RPM (measured using a counter and timer), current, voltage, and spring balance (kg). What we found was that our idealised conditions begin to drop off at higher engine speeds as we become limited by the speed of the intake and exhaust, as well as the increased effects of friction and heat loss at higher RPM. This effect was more prevalent in the petrol than the diesel engines, with the half throttle petrol engine producing the least amount of shaft power.

2 Results and Discussion

To form a relationship between the thermodynamic properties of the internal combustion engines and their engine speed, we needed a way to vary the engine speed to make the recordings. We did this by adjusting the torque supplied to the engine. Six different torque settings were used, and after each change we measured the RPM (in counts), the time taken to use 20cc of fuel, the spring balance reading (kg), as well as the current and voltage. We did this for all three engines: the half throttle petrol engine, full throttle petrol engine, and full throttle diesel engine.

We also found that the half throttle petrol engine took the longest to consume 20cc of fuel, followed by the full throttle petrol engine, and the diesel engine was the fastest⁴.

The formula for engine torque for these ICEs is

$$\tau = \text{Spring balance (kg)} \times 9.81(\text{ ms}^{-2}) \times 267\text{mm (arm length)} \quad (1)$$

Plotting this against engine speed for each of the engines (Figure 1) reveals some interesting results.

¹Spark ignition: the fuel is pre-mixed with air before entering the engine. At the point of compression, a spark is lit which ignites the fuel (V.T.O. 2013).

²Half throttle mixes half as much fuel with air as the full throttle, leading to a less powerful engine that consumes less fuel, and is more efficient at high RPM.

³In a compression ignition, only the air is compressed inside the engine. At the point of maximum compression, fuel is rapidly injected, and the high pressure causes it to immediately ignite (V.T.O. 2013).

⁴Diesel also has a higher specific heat capacity, meaning there is more energy released for a smaller amount of fuel. All engines were timed across 20cc (cubic cm) of fuel. The higher throttle petrol engine consumed fuel faster as more fuel is used in each injection.

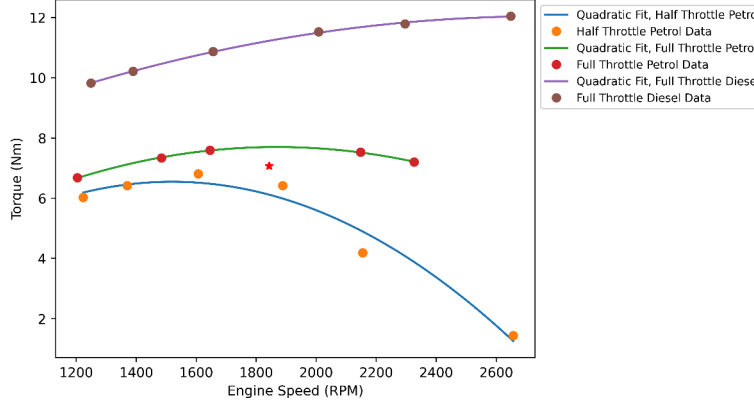


Figure 1: Plot of engine torque against engine speed for all three engine types. Data points are coloured according to the legend, and omitted outliers are labelled with red stars.

What we find is a clear quadratic relationship between torque and engine speed, characterised by a distinct 'peak' in torque followed by a decline as the engine speed increases. The reason for this, as will become a theme in subsequent results, is that our internal combustion engines are not ideal. At lower RPM, we see an expected result whereby a faster engine produces more torque. However, as the RPM gets higher the engine will approach a point where it doesn't have enough time to fill with the fuel/air mixture and is limited by the intake/exhaust. This is not taken into account in an ideal cycle. At higher speeds, we are also dealing with increased friction and more heat loss, further decreasing torque beyond this point.

Outliers were strategically chosen according to how well they fit the quadratic or linear fit, determined by approximations in Python. What we found was that some of our results produced one or two outliers in the data that could have arisen from incorrect measurements or improper engine operation. Taking into account these uncertainties, we begin to see some nonsensical fit lines that would significantly increase the uncertainty in our results. Omitting these outliers from the fitlines produced much higher correlation.

It is also clear that the full throttle diesel engine produces the most torque, followed by the full throttle petrol and half throttle petrol respectively. We will explore this more soon.

Another interesting metric is the shaft power produced by each engine. This is calculated by

$$P = \tau \text{ (Nm)} \times \omega \text{ (rad/s)}, \text{ where } \omega \text{ (Angular velocity)} = \text{RPM} \cdot \frac{2\pi}{60} \quad (2)$$

We can plot this against engine speed for each engine, as in Figure 2.

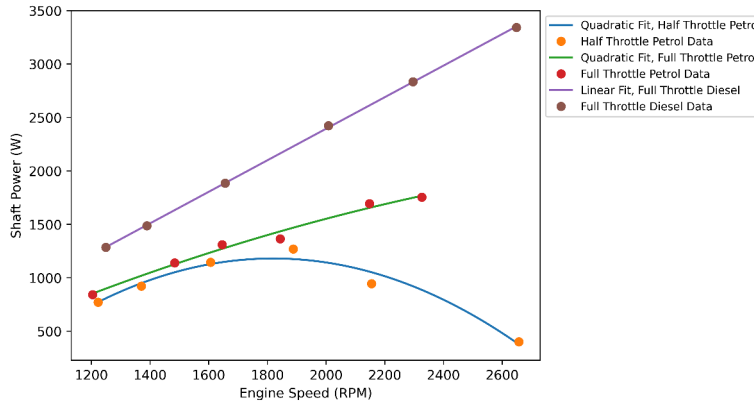


Figure 2: Plot of shaft power against engine speed for all three engine types. Data points are coloured according to the legend, and omitted outliers are labelled with red stars.

We found mostly linear correlations between shaft power and engine speed for the diesel and

full throttle petrol engines, and a quadratic relationship for the half throttle petrol engine. An inference based on this data is that, given the half throttle petrol engine is the "weakest", we see the effects of non-ideality much earlier than in the other two engines. As a result, the first two engines may also eventually peak and decline as the effects of friction, intake/exhaust speed limitations, and heat loss start to take over. Since the engines are more powerful, however, we do not see this effect at the speeds we recorded. Similar to before, the diesel engine produced the most shaft power. Interestingly, the full throttle and half throttle petrol engines produced a similar amount of shaft power in the beginning, but the half throttle engine dropped off much earlier as it uses less fuel.

The efficiency of an internal combustion engine is made up of three components; the chemical efficiency (combustion efficiency), which we are assuming to be 100%; the engine thermal efficiency; and the generator power efficiency. The engine thermal efficiency is calculated as

$$\eta_{th} = \frac{\dot{W} \text{ (Shaft power)}}{\dot{Q} \text{ (Power input (} V \times I \text{))}} \quad (3)$$

In Figure 3 this is plotted against engine speed.

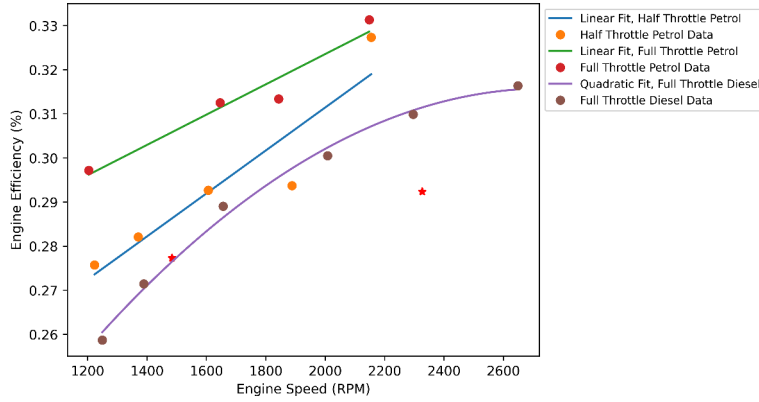


Figure 3: Plot of engine efficiency against engine speed for all three engine types. Data points are coloured according to the legend, and omitted outliers are labelled with red stars.

What we find is that the engine efficiency of the diesel engine is the lowest, then the half throttle petrol engine, and the full throttle petrol engine is the most efficient. The full throttle petrol engine should be more efficient than the half throttle. However, it is a strange result that the diesel efficiency is lower than the petrol engines. It should be higher due to the higher compression ratio and heat capacity of diesel. During the lab, the diesel engine was extremely difficult to run. This caused major uncertainties in our results that may have led to this discrepancy. Interestingly, this does not carry over into the overall efficiency, which must be due to the fact that the generator on the diesel engine was much more efficient. We will explore more uncertainties after discussing the overall efficiency.

The overall efficiency is the product of combustion, thermal, and generator efficiency, given by

$$\eta_{tot} = \eta_{com} (100\%) \times \eta_{th} \times \eta_{gen} \quad (4)$$

We can plot this against engine speed, as in Figure 4.

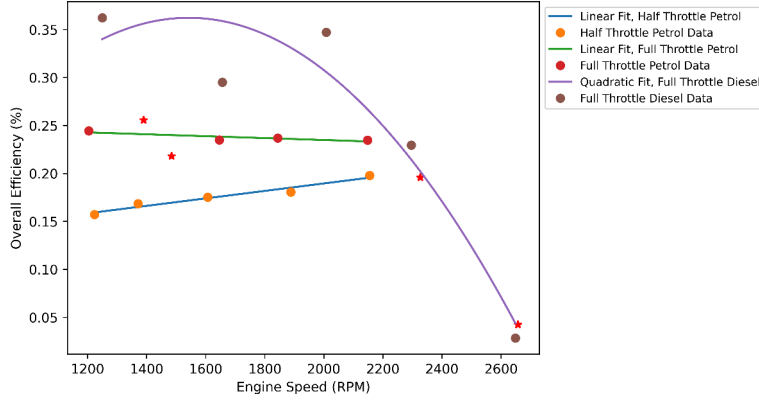


Figure 4: Plot of overall efficiency against engine speed for all three engine types. Data points are coloured according to the legend, and omitted outliers are labelled with red stars.

What we see here is that, while the diesel efficiency is the highest at lower RPM, it is the fastest to fall off at higher RPM. The full throttle petrol engine is consistently more efficient than the half throttle petrol engine, and both tend to stay seemingly consistent for the data we have taken.

This is an expected result, as diesel has the highest compression ratio and specific heat capacity, meaning it should be the most efficient engine. We see a faster falloff due to the breakdown of our ideal conditions at higher RPM, which has been explored earlier in this report. Since the engine efficiency of the diesel engine was lower, as we saw above, we can suggest that the generator on the diesel engine must be more efficient to account for this and have an overall higher efficiency. It is important to note that the power produced by the diesel engine has a very high uncertainty (about $\pm 20A$ and $\pm 20V$) compared to the petrol engines (± 3). As a result, it is much harder to infer a concrete result from this data as to which engine is more efficient. The data does seem to suggest, though, that the diesel engine is the most efficient overall.

We have made a quantitative comparison of all three engines, but it is important to infer an overall qualitative comparison, as well as compare this to a more general difference between spark and compression ignition engines. Our results show that the diesel engine is the most powerful engine. It has the highest amount of torque, produces the most shaft power, and is the least susceptible to the effects of non-ideal limitations at higher RPM. We also found the diesel engine to have the highest overall efficiency, but this is not the case for high RPM. The full throttle petrol engine is only marginally higher than the half throttle petrol engine in terms of torque and produced shaft power. The half throttle petrol engine has a much faster dropoff at higher RPM due to less fuel and more air injected. In terms of efficiency, the full throttle petrol engine is consistently more efficient than the half throttle petrol engine.

More broadly, spark ignition engines are typically less efficient and produce less power than compression ignition engines. As explored earlier, this is primarily due to the lower compression ratio of petrol (used in SI) versus diesel (in CI). Additionally, diesel has a higher specific heat capacity than petrol, meaning more heat energy is released during combustion for a smaller quantity of fuel.

3 Application

In this report we assumed a thermal combustion efficiency of 100%, meaning the entirety of the energy from the fuel being burned is converted into heat energy. In reality this may be far from the case. To quantify this energy conversion, we need to consider how much fuel we are inputting and measuring this against how much energy output we observe. The chemical energy in the fuel is given by

$$E_{\text{chem}} = \dot{m}_{\text{fuel}} \times L \text{ (Latent Heat of Vaporisation)} \quad (5)$$

We also already know the shaft power that is produced by the expansion from this heat energy. This is given in Equation 2. The combustion efficiency is

$$\eta_{com} = \frac{P_{shaft}}{E_{chem}} \quad (6)$$

We need some more information about our engine in order to collect this data. We would need a fuel flow sensor to determine the mass flow rate (an ultrasonic flow sensor, for example). This could also be determined from our results already, given we have a time and a 20cc fuel count. We also need to know the latent heat of vaporisation of both petrol and diesel.

Including this in the calculation for the overall efficiency will yield a more accurate result for the efficiency of the engines.

4 Conclusion

Our experiment effectively demonstrated the differences between spark and compression ignition in diesel and petrol engines, and we also explored how throttling an engine affects its performance. We found that the diesel engine was the most powerful engine. It produced the most torque and shaft work, and turned out to be the most efficient overall engine. We also found the petrol engines to be weaker engines and were more susceptible to the effects of non-ideal conditions at high RPM. The half throttle petrol engine was only slightly less powerful, and also less efficient than the full throttle engine. In the future, it would be beneficial to calculate the thermal efficiency of the combustion process and take this into account for the overall efficiency, as this might give us a better understanding of the efficiency of each engine. If this experiment was conducted a second time, we would take longer and record multiple recordings for each measurement. We found a number of outliers that could have been avoided, and more measurements would be needed to undertake a full uncertainty analysis involving standard errors and p-values.

5 References

Vehicle Technologies Office (V.T.O.) (2013). *Internal Combustion Engine Basics*, Energy.gov. [online]. Available at weblink.

6 Appendices

6.1 Appendix 1

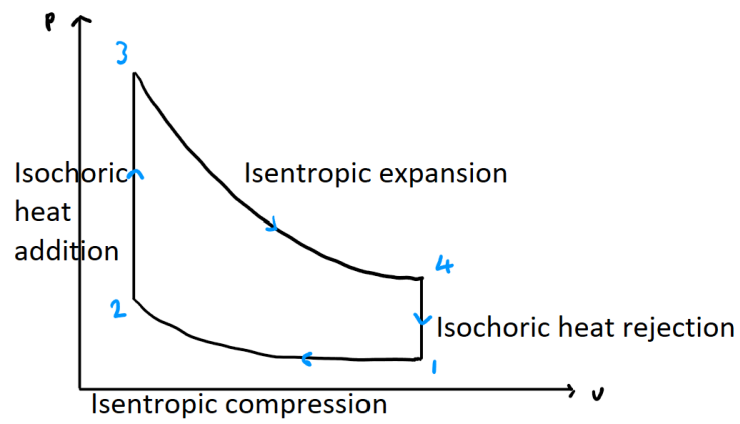


Figure 5: Idealised Otto cycle for spark ignition petrol engine.

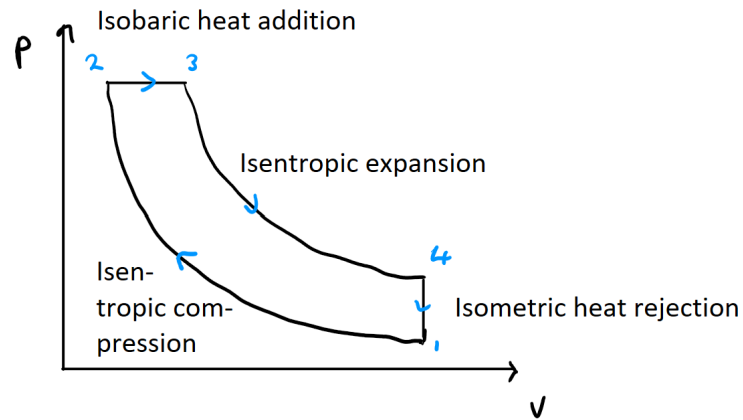


Figure 6: Idealised diesel cycle for compression ignition diesel engine.

6.2 Appendix 2

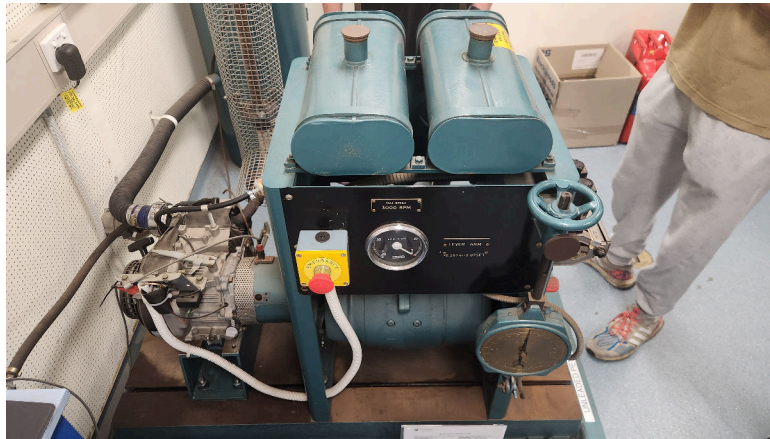


Figure 7: Image of apparatus for half throttle petrol engine. Left (pictured) is the generator. Bottom right is the spring balance, and the engine cylinder is located beneath the machine.